accuracy, and in many situations is well approximated by classical mechanics. But the conceptual distance between “a smoothly evolving wave function” and the experimental data it is meant to explain is quite large. It’s not obvious that the answer we can give to Wittgenstein’s question is the one we want. Everett’s theory might be austere in its formulation, but there’s still a good amount of work to be done to fully flesh out its implications.

In this chapter we’ll confront a major puzzle for Many-Worlds: the origin and nature of probability. The Schrödinger equation is perfectly deterministic. Why do probabilities enter at all, and why do they obey the Born rule: probabilities equal amplitudes—the complex numbers the wave function associates with each possible outcome—squared? Does it even make sense to speak of the probability of ending up on some particular branch if there will be a future version of myself on every branch?

In the textbook or Copenhagen versions of quantum mechanics, there’s no need to “derive” the Born rule for probabilities. We just plop it down there as one of the postulates of the theory. Why couldn’t we do the same thing in Many-Worlds?

The answer is that even though the rule would sound the same in both cases—“probabilities are given by the wave function squared”—their meanings are very different. The textbook version of the Born rule really is a statement about how often things happen, or how often they will happen in the future. Many-Worlds has no room for such an extra postulate; we know exactly what will happen, just from the basic rule that the wave function always obeys the Schrödinger equation. Probability in Many-Worlds is necessarily a statement about what we should *believe* and how we should *act*, not about how often things happen. And “what we should believe” isn’t something that really has a place in the postulates of a physical theory; it should be implied by them.

Moreover, as we will see, there is neither any room for an extra postulate, nor any need for one. Given the basic structure of quantum mechanics, the Born rule is natural and automatic. Since we tend to see Born rule–like behavior in nature, this should give us confidence that we’re on the right track. A framework in which an important result can be derived from more fundamental postulates should, all else being equal, be preferred to one where it needs to be separately assumed.